

TSS E-COMMERCE PLATFORM

SYSTEM OVERVIEW, PAGES ARCHITECTURE & CMS LOGISTICS

1. Executive Summary

TSS (Terminal Space Systems) is a high-fidelity, premium e-commerce web application designed specifically for creators, hardware engineers, and enthusiast gamers. The application features a distinctive, dark-themed 'cyberpunk terminal' aesthetic, deviating from generic commercial interfaces. It provides users with a futuristic, responsive console experience to browse, select, and manage cutting-edge computer systems, accessories, and industrial-grade hardware.

2. Full-Stack System Architecture

LAYER	TECHNOLOGY / SPECIFICATION
Core Frontend	React v18, Vite (fast HMR bundler), TypeScript
Styling & Theme	Vanilla CSS, HSL Custom variables, Glassmorphism, Tailwind CSS
Animations	GreenSock Animation Platform (GSAP) for premium page transitions
Icons	Lucide React (futuristic vector icons)
Core Backend	Node.js, Express framework, TypeScript
Database	MongoDB Atlas (NoSQL cloud database) with Mongoose ODM
Asset Storage	Cloudinary integration (API-driven secure image hosting)
Security & Auth	JSON Web Tokens (JWT), HTTP-Only Cookies, bcryptjs password hashing

3. Application Pages Directory

The TSS application consists of 13 primary pages, mapped using React Router client-side SPA routing:

#	PAGE NAME	ROUTE	FUNCTIONAL DESCRIPTION
1	Home	/	Main landing page showcasing featured hardware and the brand's aesthetic.
2	Laptops	/laptops	Displays available premium laptop units with custom specs and stock levels.
3	Desktops	/desktops	High-performance desktop computer workstations catalog.
4	Printers	/printers	Industrial 3D printers and high-efficiency document laser systems.
5	LED TV	/led-tv	OLED 8K cinema displays and gaming screens catalog.
6	Accessories	/accessories	Mechanical keyboards, mice, and enthusiast accessories.
7	Contact	/contact	Customer support portal featuring an interactive query terminal.
8	Warranty	/warranty	Hardware warranty registration and activation interface.
9	Downloads	/downloads	Driver updates, firmware, and software downloads page.
10	FAQ	/faq	System help desk with collapsible questions and answers.
11	Checkout	/checkout	Encrypted cart summary page and multi-step payment gateway.
12	Account	/account	User terminal for authorization, profile configuration, and order logs.
13	Admin Dashboard	/admin	Fenced CMS portal for catalog management, stock control, and analytics.

4. Administrative CMS & Analytics Console

The newly designed Admin CMS Dashboard (/admin) serves as a centralized control center. It features a premium, layered glassmorphic layout and is secured behind strict administrator role clearance. The console consists of three primary management systems:

- **Overview Analytics:**

Includes a timeframe selector (24h, 7d, 30d, 12m) that dynamically updates store KPIs (Revenue, Orders, Traffic, AOV) and renders a custom-engineered SVG Revenue Velocity Line Chart with a neon cyan glow. It also tracks category distribution and top-selling products.

- **Catalog Manager:**

Allows admins to publish new hardware or edit existing registry entries. Supports image file uploads (processed and stored via Cloudinary) or fallback URLs, categories, stock, badges, descriptions, and dynamic specification rows.

- **Fulfillment & Inventory:**

An interactive stock controller allowing admins to increase or decrease quantities in one click. Features an Order Transaction Log to manage incoming orders and update shipping statuses (Processing, Shipped, Delivered, Cancelled) in real time.

5. Deployment & Cloud Logistics

The TSS application is architected for modern cloud environments:

- Frontend Deployment: Hosted on Netlify as an optimized SPA. To handle client-side routing without 404 errors, a redirects rule (/ * /index.html 200) is deployed to prevent server-side fallback issues.
- Backend Deployment: Hosted on Railway. The backend is configured to bind dynamically to the port assigned by Railway and connect to a secure MongoDB Atlas cluster. CORS is strictly configured to only permit requests originating from the Netlify client domain.